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PHONETIC AND PHONOLOGICAL PROCESSES: THE CASE OF NASALIZATION*

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The aim of this paper is to differentiate between universal phonetic processes and language-specific phonological processes. Cross-linguistic data on coarticulatory nasalization of vowels preceding a nasal consonant were obtained across different speech rates in American English and Spanish. The data show that in American English the temporal extent of vowel nasalization varies with speech rate, whereas in Spanish, nasalization has a constant temporal extent across speech rates. It is argued that the different behavior of nasalization in these two languages reflects different inputs to speech production: In Spanish, vowels followed by a nasal are targeted as oral and nasalization is an unintended vocal tract constraint, whereas, in American English, vowels are targeted as nasalized and vowel nasalization is a phonological effect, intentionally implemented by the speaker. It is suggested that in American English, vowels followed by a tautosyllabic nasal are phonologically specified as nasal as a result of sound change. Data on perceived vowel nasalization in American English are reviewed and shown to be compatible with this proposal.

Key words: vowel nasalization, phonetic universals, coarticulation, Spanish

INTRODUCTION

Speech has been recognized to be determined (1) by "hard-wired" phonetic factors, due to universal constraints of the speech production and perceptual systems, and (2) by language-specific phonological factors programmed by the speaker. This distinction is important for speech technology: "Hard-wired" effects must be implemented in the models utilized in speech synthesis and recognition systems, while language dependent effects must be "learned" by the system. The aim of this study is to determine the phonetic (universal) or phonological (language-specific) nature of contextual vowel nasalization in Spanish and American English and to devise an empirical model that can quantify and differentiate between universal and language-specific processes. The model will be formulated on the basis of original data on coarticulatory effects of nasalization in the two languages at different speaking rates.

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Cross-linguistic studies using a variety of techniques (Clumeck, 1976; Henderson, 1984; Rochet and Rochet, 1991) show that lowering of the velum for nasal consonants necessarily overlaps the articulatory configuration of preceding vowels, and that the period of overlap varies across languages. In the present study it is hypothesized that this variation is due to the different nature of nasalization in different languages. Thus, in some languages, vowel nasalization may be a hard-wired phenomenon, mechanically linked to the presence of a nasal consonant and a by-product of the temporal organization of motor commands; whereas, in some other languages, nasalization may not be mechanical but intended, part of the linguistic organization of speech motor commands.

Cross-linguistic differences in anticipatory nasalization, lip-rounding, and vowel-to-vowel coarticulation have been explained by the principle of perceptual distinctiveness: Coarticulatory effects are governed by the phonological distinctions in a language; thus, the more distinctions there are to be maintained, the weaker are the coarticulatory effects in order to avoid perceptual confusion between distinctive and coarticulated effects (Clumeck, 1976; Lubker and Gay, 1982; Manuel and Krakow, 1984; Manuel, 1988). The cross-linguistic differences observed in anticipatory nasalization, however, cannot be explained only by the principle of perceptual distinctiveness. Coarticulatory nasalization is very limited in Japanese (Ushijima and Hirose, 1974) and Italian (Farnetani, 1986), which exhibit no apparent pressure for distinctiveness, as they only have five and seven vowel phonemes respectively, and no nasal vowels. The perceptual distinctiveness principle has been claimed to account for the very restricted coarticulatory effects of nasalization in Swedish (Clumeck, 1976), a language with a fairly crowded vowel inventory, and French (Benguerel, Hirose, Sawashima, and Ushijima, 1977; Clumeck, 1976), which has a fair number of vowel distinctions to maintain as well as distinctive nasal vowels. We propose that the similar coarticulatory behavior of nasalization in these languages with different perceptual demands is due to language-specific phonological factors, i.e., in these languages vowels followed by a nasal are targeted as oral and nasalization is the result of a physiological time constraint.

On the other hand, extensive vowel nasalization is characteristic of American English. The degree of anticipatory nasalization in American English is extreme compared to that in other languages. This suggests that it is intended by the speaker, part of the programming instructions and not a function of physiological constraints of the vocal organs. The claim that in American English anticipatory nasal coarticulation has been phonologized implies that nasalization is no longer an unintended coarticulatory effect but an intrinsic property of the vowel.

To differentiate between unintended and intended nasalization, an experiment was conducted where rate of speech (i.e., the *time* to achieve articulatory targets) was varied and its effects on the *temporal displacement* of the velum (i.e., duration of vowel nasalization) were observed for Spanish and American English. By determining whether rate of speech affected the duration of the oral or the nasalized portion of the vowel, it was possible to establish whether the vowel was articulatorily specified as oral (with mechanical nasalization) or as nasalized. The rationale of the experiment was the following: Given that speech rate is a higher level adjustment, then if the vowel is targeted

as nasalized, and therefore nasalization is part of higher level instructions, nasalization should adjust to rate variations. If, on the contrary, the vowel is targeted as oral and nasalization is due to low level vocal tract constraints, the temporal extent of nasalization should not vary with speech rate.

EXPERIMENT

Method

Three speakers of American English (two males and one female) and three speakers of continental Spanish (two females and one male), with no obvious speech or hearing defects, read a randomized word list consisting of all possible combinations of $C_1V_1'V_2C_2$, where C_1 = t, n; V_1 = i, ɑ; V_2 = i, e, ɑ; C_2 = t, n. High and low vowels were chosen since vowel height has been shown to have an effect on velum lowering (Moll, 1962; Lubker, 1968; Fritzell, 1969; Ohala, 1971; Ushijima and Sawashima, 1972; Bell-Berti, Baer, Harris, and Niimi, 1979; Henderson, 1984). The carrier sentence for English speakers was "Guess __ soon". The Spanish carrier sentence was "Dos __ son" (*"They are two __"*). The subjects were asked to read the 24 test sentences twice at five different speech rates: (1) overarticulated, overly slow speech ("as if talking to a deaf person who was lip reading"); (2) careful, slow speech ("as if reading out loud to a formal audience in a big lecture hall"); (3) normal conversational speech; (4) fast speech; (5) underarticulated, overly fast speech ("as fast as you possibly can"). The four speech rates most equidistant in tempo were analyzed for every speaker. The vowel sequences were pronounced in a comparable way by the English and Spanish speakers, with no intervening glottal stop. Only the American speaker AV showed an intervening glottal stop in the sequences [iV] at the slower speech rate, and these sequences were discarded.

To track the time-varying positions of the velum, a nasograph (see Ohala, 1971, for a description) was inserted into the subjects' nasal cavity and pharynx, and the traces of velopharyngeal port opening/closing and of the acoustic waveform were obtained on a Siemens Oscillomink chart recording device in the standard way (Ohala, 1971; Clumeck, 1976). To check the positioning of the nasograph and calibrate the movements of the velum, speakers were asked to produce before and after the experiment a sustained [s] and a sustained [n] in alternation. The signals obtained during these test sounds were monitored with an oscilloscope and recorded on FM tape for further analysis. The sustained [s] is associated with maximum velum height, whereas the sustained [n] is associated with minimum velum height. These extreme positions allowed us to check that the nasograph was tracking the full movements of the velum in the test utterances of the experiment, and they were used to calibrate the recorded velum signals. The nasograph trace is a reliable indicator of the timing of velopharyngeal port opening, but it is not reliable for the spatial displacement of the velum across different speakers, since, depending upon the positioning of the nasograph, the magnitude of light impinging on the sensor might differ. Thus, the relative magnitude of velum opening can be compared within data from one speaker but not across different speakers. Measurements of vowel duration and timing of soft palate lowering before nasal consonants were done

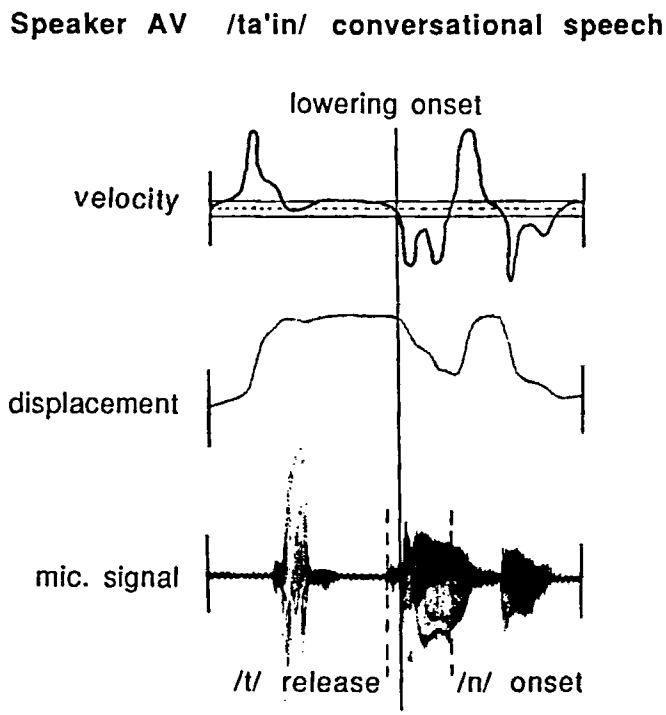

Fig. 1. Traces for the velocity function of the velum (with the defined noise band), velic displacement, and acoustic waveform. Velar lowering begins when the velocity function crosses the noise band. Acoustic landmarks are labelled.

in [t^hV'V_n] sequences. Measurements of vowel duration were obtained (1) for the aspiration period [h], (2) for V_1, and (3) for V_2. Measurements were done by hand on Oscillomink traces.

Analysis procedure

The method used for determining the onset of velum lowering was to consider movement to begin at the time when the velocity function for the velum crossed a noise band around zero. The velocity noise band was defined as 10% of the peak velocities of the velar movement gestures for each speaker (Krakow, 1989). Figure 1 illustrates the procedure for onset determination. It shows the velocity function of the velum with the defined noise band, velic displacement, the acoustic waveform, and the point in time at which onset of velum lowering is considered to begin.

Determination of onset of velum lowering in two-stage and multi-stage velar gestures

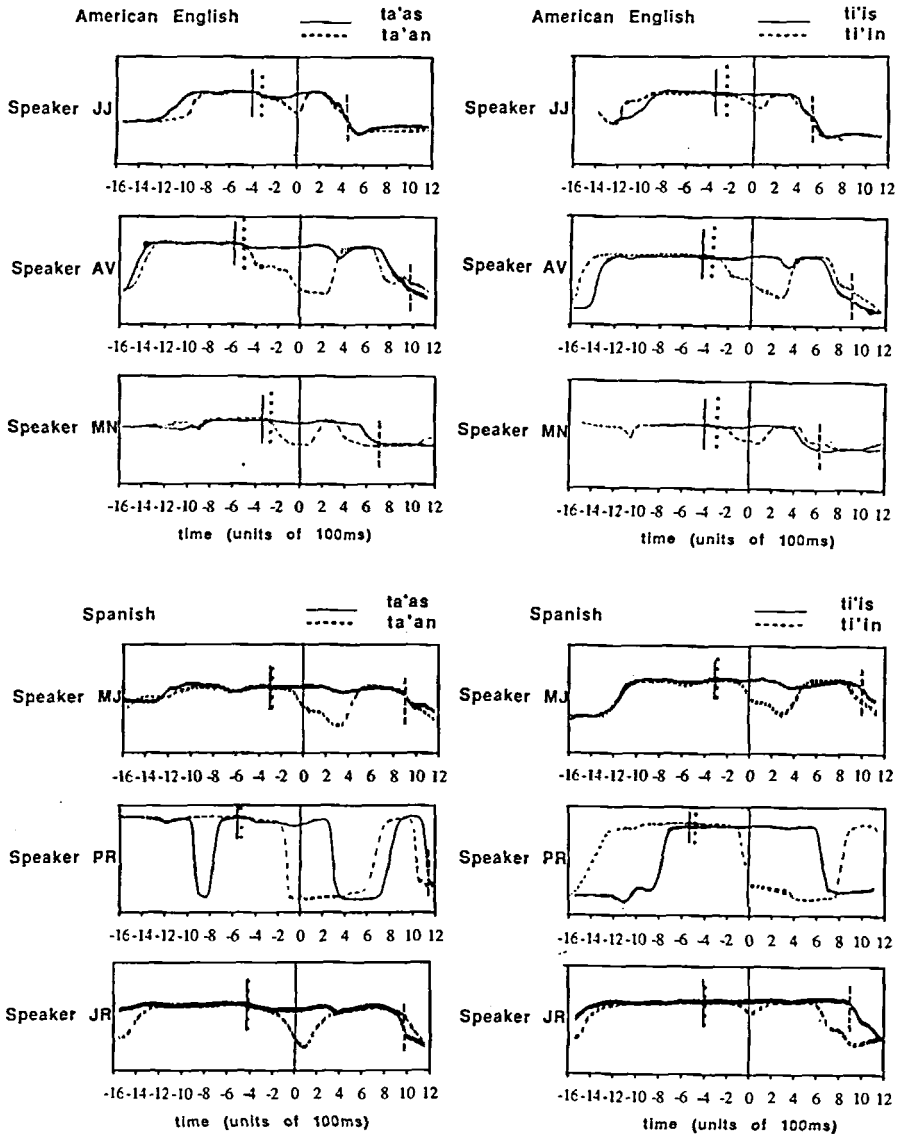

Fig. 2. Velum displacement in oral and nasal sequences for American English and Spanish subjects. The sequences were spoken at the slower speaking rate. Time is represented along the abscissa, with 0 marking the onset of the nasal consonant. The release of the /t/ is indicated by a solid line. The beginning of voicing for the vowel is indicated by a dotted line. The onset of the /n/ in the carrier phrase is indicated in the oral sequences by a broken line. In the oral sequences, speaker PR shows an important velar lowering before the onset of /n/ in the carrier phrase, which is associated with a pause made between the two /s/ sounds in

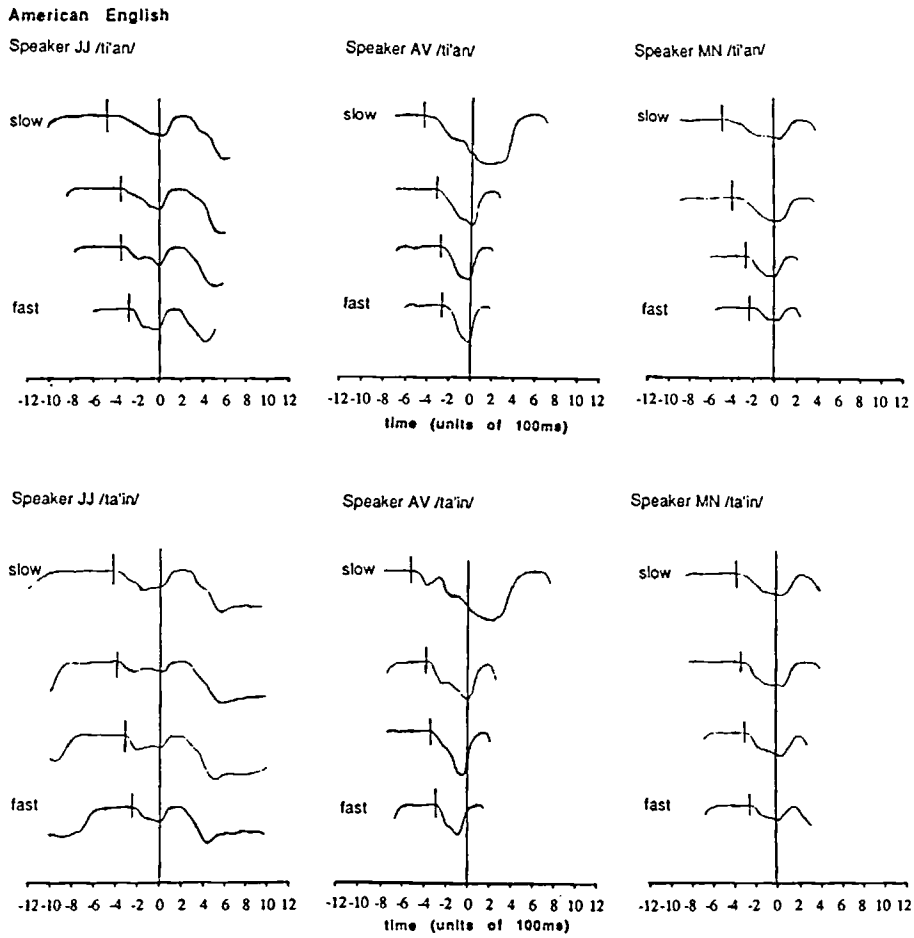

Fig. 3. Displacement functions of nasal utterances spoken at four different speaking rates for all three American English subjects. Time is represented along the abscissa, with 0 marking the onset of the /n/. The release of the /t/ is indicated.

– usually those at slower speaking rates in American English – is more controversial. Bell-Berti and Krakow (1991) claim that the first lowering movement in CV_nN sequences (where V_n stands for any number of vowels) in slow speech corresponds to the intrinsic velar gesture for vowels in a strictly oral context (oral vowels involve lower velum positions than oral consonants) and is unrelated to anticipatory nasalization. Contrary to this claim, our data show that, in American English, the magnitude of the velar lowering movement for vowels in a nasal context is larger than the one found in strictly oral contexts. Figure 2, which shows CV'VC and CV'VN minimal pairs in American

English and Spanish, allows us to compare the velar lowering gesture associated with the vowel in a strictly oral context with that for the vowel followed by a nasal for each speaker. Note that the nasograph signal is a reliable indicator of relative spatial displacement of the velum within each speaker's range. Figure 2 shows little or no velar lowering for oral vowels in an oral context (solid trace), both for American and Spanish speakers. The velocity function for the velar lowering movement for vowels in an oral context does not go beyond the defined noise band (except in some cases involving V_1 = ɑ for the American English speaker JJ with a heavily nasalized speech) and, consequently, intrinsic vowel lowering associated with the CV transition was not included in the measurements of anticipatory nasalization. Figure 2 shows that, in American English, the magnitude of the velar lowering movement for vowels in a nasal context (dotted trace) is not attributable to intrinsic vowel lowering for oral vowels (as Bell-Berti and Krakow, 1991, claim) but to vowel nasalization (whether intrinsic, as we want to claim, or anticipatory).

Consequently, the criterion utilized to determine the onset of velar lowering in complex gestures was the same as for one-stage gestures. When the velocity function for the first lowering movement exceeded the noise band – as was the case for vowels in a nasal context – velar movement was considered to have begun.

Figure 3 shows the traces for velar movements in American English for the sequences [ti'ɑn] and [tɑ'in] at different speaking rates (1 = slowest, 4 = fastest). Note that complex gestures are found at the slower speaking rate (as predicted by Bell-Berti and Krakow, 1991) for all speakers in [ti'ɑn] sequences and for speakers JJ and AV in [tɑ'in] sequences. However, complex gestures are never found in the Spanish data, as illustrated in Figure 4, which shows the traces for the equivalent sequences for Spanish. (The small lowering after the [t^h] release exhibited by speaker PR in [tɑ'in] sequences is associated with the intrinsically lower velum position for the vowel than for the preceding oral consonant – compare the oral sequence for the same speaker in Figure 2 – and is unrelated to coarticulatory nasalization.) This cross-linguistic difference suggests that the lowering movements associated with the vowel in a nasal context are a language-specific feature of American English.

Results

The results of the measurements for the three American English speakers are presented in Figure 5, which shows the mean duration in milliseconds of the oral and nasalized portion of the vowel sequence – including the aspiration period – for [iV] and [ɑV] sequences. The onset of velic lowering is marked 0 on the abscissa, and segments appearing to the right of 0 are the nasalized portions of the vowel sequence. Varying speech rates appear on the ordinate. Speech rate was plotted by determining the average duration for the vowel sequences.

Figure 5 shows that in American English the vowel is nasalized throughout except for the non-nasalized portion which corresponds to the aspiration period as obtained from the acoustic waveform (mean non-nasalized portion for speaker JJ = 59.9 msec, AV = 52.3 msec, MN = 49.7 msec; mean aspiration period for speaker JJ = 58.3 msec, AV = 61.5 msec, MN = 49.8 msec). Furthermore, in some cases velic lowering precedes

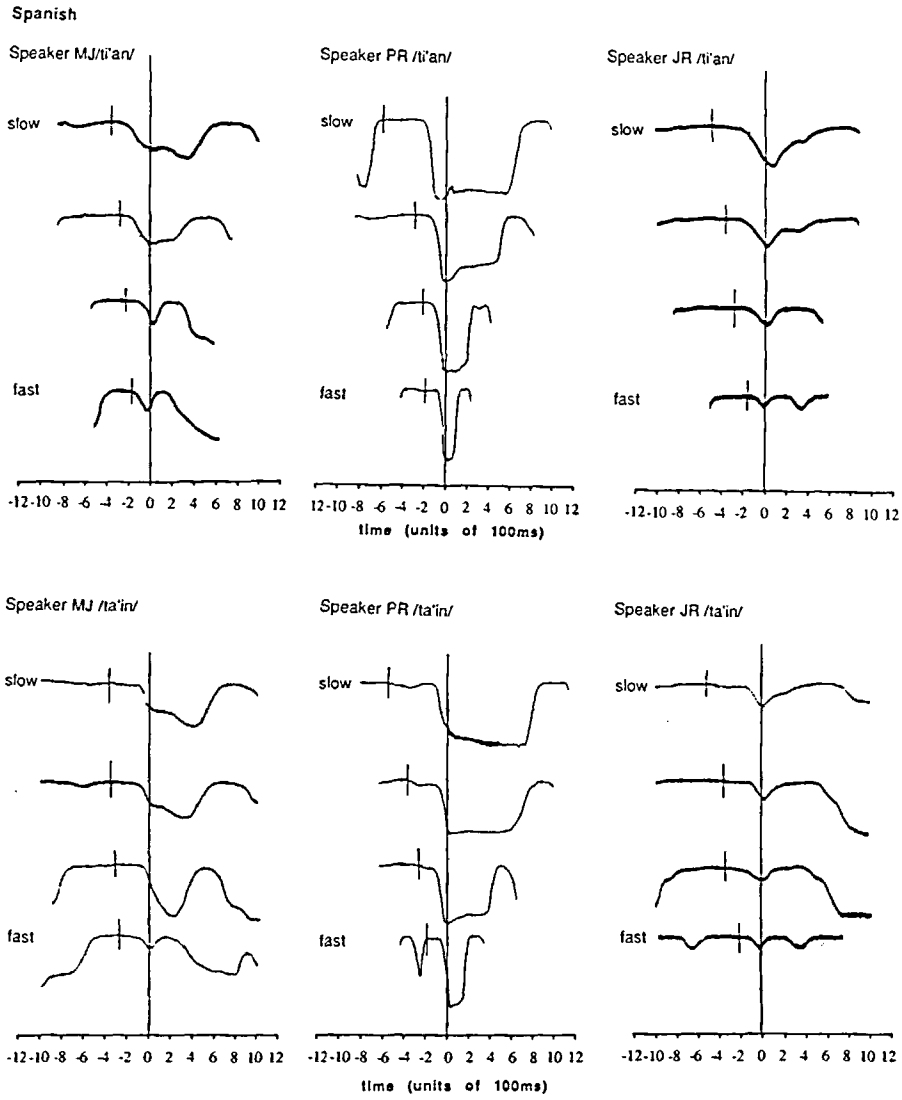

Fig. 4. Displacement functions of nasal utterances spoken at four different speaking rates for all three Spanish subjects. Time is represented along the abscissa, with 0 marking the onset of the /n/. The release of the /t/ is indicated.

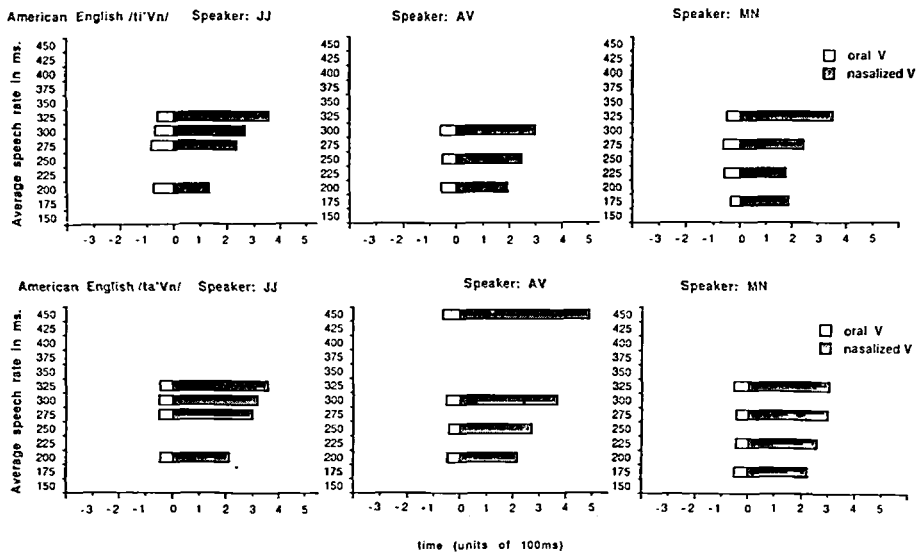

Fig. 5. Mean duration in msec of the oral and nasalized portions of the vowel sequence in /ti'Vn/ and /ta'Vn/ (including aspiration) for American English subjects on the abscissa. Onset of velic lowering is marked from 0. Average speech rate in msec appears on the ordinate.

the onset of voicing for the vowel, resulting in a nasalized aspiration, [ɦ]. This indicates that in American English the voiced portion of the vowel sequence is completely nasalized. As shown in Figure 5, nasalization in American English *varies with speech rate* (the faster/slower the speech rate, the shorter/longer the nasalized portion) with some saturation (as evidenced in the slower speech rate for speaker MN in [aV] sequences) and incompressibility effects (illustrated in the fastest speech rate for speaker MN in [iV] sequences). Figure 5 shows that in American English the velum begins lowering approximately at the onset of voicing for the vowel. Since vowels have different durations across speech rates, nasalization begins at different points in time before the onset of the nasal consonant; earlier for slower speaking rates and later for faster speaking rates, with the result that the voiced portion of the vowel across all rates has *the same percentage of nasalization* (100%), despite the durational differences.

Figure 6 shows the results for the three Spanish speakers for [iV] and [aV] sequences. It can be observed that the nasalization period in Spanish shows a roughly constant duration across different speech rates. Only at the fastest speech rates do some speakers (MJ [iV]; PR [iV]; JR [iV], [aV]) succeed in reducing the nasalized portion, which indicates that under unusually fast speech conditions speakers may increase the velocity of the velic lowering movement, thus increasing articulatory cost (Nelson, Perkell, and Westbury, 1984; Kelso, Vatikiotis-Bateson, Saltzman, and Kay, 1985). The reduction of

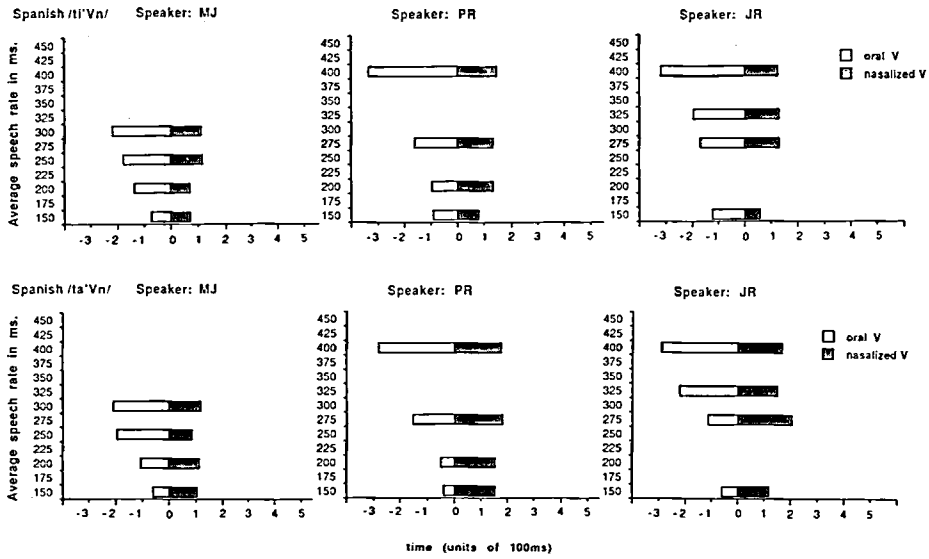

Fig. 6. Mean duration in msec of the oral and nasalized portions of the vowel sequence in /ti'Vn/ and /ta'Vn/ (including aspiration) for Spanish subjects on the abscissa. Onset of velic lowering is marked from 0. Average speech rate in msec appears on the ordinate.

the nasalized portion in fast speech might be interpreted as a carry-over effect: The onset of velar lowering is delayed by the high velum requirement of the vowel.

Figure 6 shows a longer nasalization period for [aV] sequences than for [iV] sequences. It seems reasonable to suggest that nasalization has a constant duration in both cases and that differences are due to further low-level adjustments due to vowel height. The fact that for [iV] sequences the constant nasalized period (k) across speech rates for the different speakers (MJ, k = 91.6 msec; PR, k = 124.3 msec; JR, k = 97.6 msec) is longer than the minimum transitional period towards a nasal consonant configuration (40 msec for [ii] sequences) is due to the fact that the computed k value is the mean of the values for different V_2 vowels. This indicates that the height of V_2 also has an effect on velum lowering; this effect is presently under study. The effect of V_2 is less evident for [aV] sequences. To sum up, the results in Figure 6 suggest that nasalization has a *constant value* across speech rates, and that the oral portion of the vowel varies as a function of speech rate. Thus, Spanish vowels can be said to be targeted as oral, and nasalization is the result of a physiological time constraint.

The data indicate that the percentage of vowel nasalization in Spanish is variable. Figure 6 shows that velic lowering begins at roughly the same point in time before the onset of the nasal consonant across different speech rates, thus resulting in a *different percentage of vowel nasalization* across speech rates (vowels in faster speech rates being

more nasalized because they are shorter). These results are in agreement with the results of Clumeck (1976, Table V) for Swedish, where he found that velum lowering occurs at a roughly constant time before the onset of the nasal consonant regardless of whether the preceding vowel is long or short, thus resulting in short vowels being more heavily nasalized than long vowels. These results also resemble the data of Farnetani (1986) for Italian, of Ushijima and Hirose (1974) for Japanese, and of Benguerel *et al.* (1977) for French, which all show a temporally restricted coarticulatory nasalization.

The strikingly consistent behavior shown by the different speakers within each language group indicates that there is a principled difference in the targeting of nasalization in each language.

DISCUSSION

The finding that nasalization in Spanish occupies a constant number of milliseconds across speech rates indicates that it may take a limited and stable time to establish the articulatory configuration for the nasal consonant. Thus, nasalization in Spanish can be considered as an unintended effect which is independent of higher level adjustments such as speaking rate. If speakers were using vowel nasalization distinctively, one would expect that the nasalized portion in one rate of speech would differ from that in another rate by an amount proportional to the difference of the duration of the vowels, so as to maintain a constant perceptual distance across speech rates. This is the case for American English. Thus, it can be deduced that in American English nasalization does not occur automatically but that it is present in the phonological specification of the segment, i.e., part of the programming instructions of the language.

Co-production models of coarticulation that do not include language-specific organization of articulatory gestures cannot account for the data obtained. These models predict that anticipatory nasal coarticulation starts at a stable time before the nasal consonant and thus cannot account for nasalization present during the acoustic duration of the vowel in American English. This is due to their failure to distinguish between nasalization as a property of the vowel and as a coarticulatory effect.

Feature-spreading models could account for the data obtained by invoking differences in feature specification for Spanish and American English. Thus, in American English the vowel would be unspecified (i.e., neutral) for nasality, and interpolation between the oral target for [t] and the nasal target for [n] would result in a vowel that is thoroughly nasalized. Spanish vowels, on the other hand, would be specified for this feature as [–nasal] and would have a target for nasality similar to that of [t]. Interpolation would then take place between the vowel and the nasal, and it would affect minimally the orality of the vowel. However, stating that the vowel is neutral or non-neutral for a particular feature is just relabelling the facts and does not provide an *explanation* for *why* the vowel is or is not nasalized.

The explanation we propose is that, in American English, vowels followed by a nasal in the same syllable are phonologically specified as [+nasal] as a result of sound change. Thus, in this language vowel nasalization is not the result of anticipatory coarticulation

but an intrinsic property of the vowel. Most likely, the origin of vowel nasalization in American English lies in the transition time towards a nasal consonant configuration. Some 50 to 70 msec of the portion of the vowel adjacent to a nasal consonant is nasalized due to the inertial properties of the velum. The intrinsically lower velum position for oral vowels than for oral consonants (which seems to be more extreme in American English *vis-à-vis* other languages) may have contributed to nasal coupling during the acoustic duration of the vowel. Listeners may have reinterpreted the resulting nasalization as an intended feature of pronunciation and may have incorporated and exaggerated it in their own speech. Thus, vowel nasalization may be the result of a sound change which has become part of the language-specific instructions of American English. This interpretation is supported by the fact that, at slower speech rates, where the time constraint is not present and the gestures for the vowel and for anticipatory nasalization could show up separately, some speakers show a single, sharp velar lowering movement at the beginning of the vowel (Figure 3, speakers AV and MN in the sequence /tiˈɑn/ at speech rate 2; speaker MN in the sequence /tɑˈin/ at the slower speech rate; all three speakers in /tɑˈin/ at speech rate 2), and those speakers that present multistage gestures show a lower velum position in /VN/ sequences than in oral sequences (Figure 2), which indicates that the feature is exaggerated when produced intentionally.

The cross-linguistic data examined suggest that the time-varying properties of nasalization cannot be accounted for by coarticulatory models that do not include language-specific organization of gestures, nor can they be explained only in terms of preserving perceptual distinctiveness. The explanation provided in this paper is that language-specific control of coarticulation can be explained in terms of attaining the articulatory and acoustic phonological requirements of the segment as oral or nasalized, i.e., attaining the phonetic quality which is the norm in a given language variety.

Perceptual implications

Most views of speech production and speech perception rest on the assumption that the encoding units utilized in producing and perceiving speech are discrete segments which have different target values for phonetic properties. We will first review the available perceptual data on nasalization in terms of segmental approaches and then in terms of a diphone model, i.e., a model that assumes that phonetic properties are mediated by phonological units of diphone size.

It is a well known fact that knowledge of how phonetic units are produced influences the perception of speech. Thus, if American English speakers produce vowel nasalization in /VN/ sequences intentionally, as is proposed here, then listeners must be presumed to perceive vowel nasalization as an intrinsic property of the vowel and not as an induced effect of the following nasal. Perceptual data, at first sight, seem not to support this prediction. Data on perceived vowel nasalization (Kawasaki, 1986; Krakow and Beddor, 1991) show that American English listeners perceive nasal vowels in a nasal context as less nasal than the same vowels in an oral context, i.e., they factor out nasality from vowels in a nasal context. This indicates that, if a nasal consonant is present, listeners do not attribute nasality to the vowel, as predicted above, but to the nasal consonant and, thus, subtract it from the vowel. However, two points have to be made regarding

the perceptual data. In the first place, data on perceived nasalization show that speakers *may* factor out nasalization from the vowel, but they do not *always* factor it out (i.e., they do not always attribute it to the context and subtract it from the vowel). In Kawasaki's (1986) Experiment 1, where speakers were asked to judge degree of nasality on a five-point scale, she found that half of the subjects did not factor out nasality. Similarly, Krakow and Beddor (1991) found that in a majority of the cases listeners did not subtract nasality as a contextual effect. This indicates that speakers follow different strategies in judging ambiguities in the acoustic signal (coarticulatory *vs.* intended nasalization) with respect to the speaker's intentions. Speakers who do not factor out nasality as a contextual effect but identify it as a property of the vowel lend support to our interpretation that nasalization in American English has been phonologized and is an intrinsic feature of the vowel.

In the second place, findings that listeners factor out or "correct" nasalization as an unintended coarticulatory effect can be viewed as evidence of perceptual dissimilation or hypercorrection (Ohala, 1981; Ohala, in press b). In fact, the difference between correction and hypercorrection lies in whether the effect is unintended (and, therefore, adequately corrected or factored out by the listener) or intended by the speaker (and, consequently, overcorrected by the listener). If one posits a phonological representation of the type of /$\widetilde{V}$N/, with the vowel and the nasal intentionally nasalized and thus sharing the same feature [+nasal], the listener may identify nasality in the consonant but may erroneously subtract nasalization from the vowel as an expected coarticulatory effect of the nasal consonant. The difficulty of distinguishing between phonological and coarticulatory nasalization might account for the listeners' dissimilatory perceptual behavior in identifying intended nasalization as context determined. This difficulty is evidenced by Krakow and Beddor's (1991) results which show that American English listeners are uncertain as to the nasality of vowels in nasal contexts, i.e., they cannot distinguish whether nasalization is a property of the vowel or a coarticulatory effect. Also, the fact that distinctive nasal vowels do not contrast with oral vowels when adjacent to a nasal consonant (Ferguson, 1974; Kawasaki, 1986) shows the difficulty of distinguishing between intrinsic and contextual nasalization in a nasal context. If, on the contrary, the nasal consonant is deleted in American English, as is usually the case before homorganic voiceless stops (Malécot, 1960), the speakers perceive vowel nasalization, as shown by the fact that they contrast minimal pairs such as *cat* vs. *can't*, *set* vs. *sent*. Since the perceptual data available on American English can be interpreted as perceptual dissimilation, they are compatible with our proposal that American English vowels followed by a tautosyllabic nasal are intentionally nasalized.

The encoding units of speech production may not be discrete segments as we have assumed so far, but larger units such as diphones (or syllables) which may have the articulatory gesture as the functional unit of motor action. This proposal has been made by several authors, both for speech production (Kozhevnikov and Chistovich, 1965; Ohala, in press a) and speech perception (Whalen, 1989; Nearey, 1990). A diphone model that incorporates contextual variation into larger, more nearly invariant syllabic units can account more adequately for contextual effects. The articulatory goals might then be the transitions of sounds (diphonemes) with the described trajectories towards that

goal, which might change from language to language. According to this view, the articulatory trajectories towards a /VN/ transition would be different for Spanish and American English. Vowel nasalization in American English would be the result of the articulatory trajectory phonologically specified for that language, and not a property of the vowel itself as in a segmental approach. In languages such as Spanish, orality in the vowel would result from the articulatory trajectory phonologically specified for that language plus a mechanical coarticulatory effect. This approach would entail that coarticulation within diphones, syllables or encoding units is programmed, while coarticulation across units is due to mechanical accommodation to surrounding sounds. The stronger coarticulatory effects of nasalization found within syllables than across syllables (Ohala, 1971; Krakow 1989) seem to support this interpretation.

To sum up, data on perceived nasalization are compatible with the proposal that, in American English, vowels in a nasal context are phonologically specified as nasal.

CONCLUSION

Cross-language differences in vowel nasalization can be explained in terms of language-specific phonological requirements. The experimental data presented show that vowel nasalization in Spanish takes place at a stable time before the nasal consonant across different speech rates. Thus vowel nasalization in Spanish can be considered a low-level coarticulatory effect which is independent of higher level adjustments such as speaking rate. Spanish vowels are consequently phonologically specified as oral with mechanical nasalization. In American English, on the other hand, nasalization is present throughout the duration of the vowel across different speaking rates. Thus, nasalization is a higher-level or intended effect which adjusts to speech rate variations. Consequently, in American English vowels followed by a tautosyllabic nasal are phonologically specified as nasalized. Finally, it is suggested that vowel nasalization in American English is the result of sound change.

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